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Trends and Cloud Services Overview

1. Advantages of Serverless Architecture

Serverless computing removes the burden of managing infrastructure, allowing developers to concentrate solely on application logic. Services like AWS Lambda and Azure Functions handle automatic scaling based on workload, resulting in greater efficiency and lower costs. This model supports rapid development and quicker deployment times. Additionally, serverless architecture helps reduce infrastructure overhead and operational complexity, making it a great fit for fast-paced, innovative environments.

2. Progressive Web Applications (PWAs)

Progressive Web Apps are a hybrid between traditional web applications and mobile apps. They function across various devices, whether on desktop or mobile, and do not require installation from app stores. PWAs feature quick loading times, offline accessibility, and high performance, enhancing user engagement. With capabilities such as push notifications and home screen shortcuts, they deliver an app-like experience directly through the browser.

3. Impact of Artificial Intelligence and Machine Learning on Software Architecture

Artificial Intelligence (AI) and Machine Learning (ML) are revolutionizing software architecture by enabling applications to process data intelligently, adapt to user behaviour, and make informed predictions. These technologies contribute to personalization, automation, and improved user interactions. The integration of AI/ML into system architecture leads to smarter functionality, enhanced analytics, and optimized workflows.

Cloud Computing Service Models

Cloud computing is categorized into three fundamental service models:

- **Software as a Service (SaaS):** Applications are delivered over the internet, eliminating the need for local installation (e.g., Google Workspace). It’s mainly used by end-users.
- **Platform as a Service (PaaS):** Developers can build, deploy, and manage applications without managing the hardware and infrastructure (e.g., Google App Engine).
- **Infrastructure as a Service (IaaS):** Offers virtualized infrastructure resources like compute power, storage, and networking over the internet (e.g., AWS EC2), ideal for companies that require flexible and scalable systems.